CYCLON convergence: proved cases and progress toward a general theorem

Research working note

19 October 2026

Abstract

We collect the current results in one self-contained source. The main positive result proves convergence of mandatory-removal CYCLON with shuffle length one, arbitrary network size and cache capacity, independent uniform activations, and arbitrary finite initial ages. A constructive common-atom argument gives a unique stationary law, an explicit uniform geometric total-variation bound, and uniform local samples conditional on availability. At capacity one the stationary law is explicitly uniform on zero-age rooted spanning trees. We also prove convergence for three peers with capacity and shuffle length two. For every shuffle length we prove accessibility of complete initial-lineage erasure, with an age-dependent path bound; this does not yet establish recurrence or mixing. Controlled replacement and numerical-age routing lemmas support these results.

Earlier results are retained: a capacity-one retention theorem and counterexamples for particular scheduling and retention conventions. Each statement specifies its assumptions. In particular, the new positive theorems use independent uniform activations rather than the fixed periodic phases in the original paper. A convergence proof for arbitrary network size, capacity, and shuffle length remains open here.

1 Scope and exact protocol conventions

The motivating target is a local total-variation guarantee of the kind proved for PEERSWAP in [1]. Its interchange-process reduction uses a fixed underlying graph and does not automatically extend to partial cache exchanges.

We consider a static membership $V = \{1, \dots, n\}$, reliable atomic exchanges, and caches without self entries or duplicate identities. Descriptors contain an identity and a nonnegative integer age. Enhanced CYCLON increments the initiator's cached ages, selects an oldest descriptor as partner, sends a fresh self descriptor in place of that partner along with other selected descriptors, and receives a subset of the partner's cache [2].

The update convention matters. We distinguish:

- *Retention:* discard self and already-known incoming descriptors, and replace sent cache entries only when a usable incoming descriptor is inserted. A contacted partner is retained when no replacement is usable. Discarded duplicates do not refresh ages.
- *Mandatory removal:* unconditionally remove the contacted partner, even when the response yields no usable replacement.

These conventions differ on reciprocal exchanges. The published paper's seven steps are followed by the explicit statement that the contacted peer disappears [2, Section 2.2]. We therefore use mandatory removal as the baseline for the new positive results. The retention results below concern the separately specified alternative; their obstructions do not establish failure of mandatory-removal CYCLON. Underfull buffers, replacement choices, and scheduling are specified explicitly wherever needed.

Time counts global activation events. Periodic phases and independent uniform random activations are treated separately. For a cache that can supply b entries, $S_i^{(b)}$ is a uniformly selected, uniformly ordered b-tuple of distinct cached identities. When a requested tuple is unavailable, write $\perp$. Uniformity at one peer and time does not imply independence across peers or times.

2 A periodic counterexample under either convention

At $c = \ell = 1$, if the initiator is p, its neighbor is q, and q's cached identity is r, with p, q, r distinct, the unique exchange outcome is

$$(f(p), f(q)) = (q, r) \quad \longmapsto \quad (f'(p), f'(q)) = (r, p). \tag{1}$$

All other identities stay fixed. Both update conventions agree here.

Theorem 1 (Failure with fixed periodic phases)**.** *There is a four-peer enhanced-CYCLON execution with a strongly connected initial overlay, full caches, $c = \ell = 1$, and one activation per peer per round, in which a peer's sample never converges to uniform. No exchange has a tie, unusable replacement, or empty cache.*

Proof. Activate peers in order $1, 2, 3, 4$ in every round, using distinct fixed phases within a common period. Initialize all descriptor ages to zero and the identity state to $(3, 4, 2, 1)$. Equation (1) gives

After activation	$f(1)$	$f(2)$	$f(3)$	$f(4)$
Initial	3	4	2	1
1	2	4	1	1
2	2	1	1	2
3	3	1	2	2
4	3	4	2	1

The initial graph is the directed cycle $1 \to 3 \to 2 \to 4 \to 1$. In every exchange p, q, r are distinct. The initiator's sole descriptor is incremented from zero to one, then discarded and replaced in its outgoing message by its own age-zero descriptor. The responder's age-zero descriptor moves to the initiator. Thus all stored ages are again zero after each exchange. The entire state and scheduler phase repeat after four activations.

At each round boundary, peer 1 returns peer 3 deterministically. For all $k \geq 0$,

$$\left\| \mathcal{L}(S_1^{(1)}(4k)) - \mathrm{Unif}\{2, 3, 4\} \right\|_{\mathrm{TV}} = \frac{2}{3}.$$

Moreover peer 1 never stores peer 4, even between round boundaries. Every mixture over observation times therefore remains at least $1/3$ from uniform. Time averaging cannot repair this execution. □

3 Frozen multi-entry states with random scheduling

This section uses retention and discards already-known descriptors *without refreshing their ages.* It allows random initiators, random subsets, and any tie-breaking rule.

Theorem 2 (Strongly connected frozen states)**.** *For every even $n \geq 4$, every $2 \leq c \leq n - 2$, and every $1 \leq \ell \leq c$, there is a strongly connected, full-cache initial state for which all cache identities remain fixed forever under enhanced CYCLON with retention. Every peer always has*

a unique oldest descriptor. This holds for every activation sequence and every outcome of the protocol's random subset selections.

For every $1 \le b \le c$ and every observer i, its sample distance is the positive constant

$$\left\| \mathcal{L}(S_i^{(b)}(t)) - \mathrm{Unif}((V \setminus \{i\})_b) \right\|_{\mathrm{TV}} = 1 - \frac{(c)_b}{(n-1)_b}, \qquad t \ge 0, \tag{2}$$

where $(a)_b = a(a-1)\cdots(a-b+1)$.

Proof. Partition peers into pairs (p_k, q_k), $1 \le k \le m = n/2$. For each pair choose a set S_k of $c-1$ identities outside the pair that contains p_{k+1}, with indices modulo m. This is possible since $m \ge 2$ and $c - 1 \le n - 3$. Set

$$C_{p_k} = \{q_k\} \cup S_k, \qquad C_{q_k} = \{p_k\} \cup S_k.$$

Give the reciprocal partner descriptor age 1 and every other descriptor age 0. Each pair is internally mutually reachable; the edges to p_{k+1} connect the pairs around a directed ring. Consequently the overlay is strongly connected.

Suppose p_k initiates. Its unique oldest descriptor is q_k. The outgoing message consists of its own fresh identity p_k and $\ell - 1$ elements of S_k, all already in q_k's cache. The responder therefore discards every incoming descriptor. Every descriptor it returns is either p_k, which is a self entry at the initiator, or an element of S_k, already known there. The initiator also discards every incoming descriptor. No cache identity and no stored age is overwritten.

Incrementing all ages at the initiating cache preserves its strict age difference between the partner and the common neighbors. Passive responses do not increment ages. The same argument applies when q_k initiates. Induction proves that the identities remain frozen and the designated partners remain uniquely oldest for all time.

The output is uniform on the $(c)_b$ ordered tuples contained in the fixed cache, whereas the target is uniform on all $(n-1)_b$ tuples. For uniform laws on a subset A of a finite set B, their total-variation distance is $1 - |A|/|B|$, proving (2). □

The smallest example, with $n = 4, c = 2$, is

$$C_1 = \{2^{[1]}, 3^{[0]}\}, \quad C_2 = \{1^{[1]}, 3^{[0]}\},$$
$$C_3 = \{4^{[1]}, 1^{[0]}\}, \quad C_4 = \{3^{[1]}, 1^{[0]}\},$$

where brackets denote ages. It works for both $\ell = 1$ and $\ell = 2$. Peer 1 never samples peer 4; its singleton distance is $1/3$. For a realistic illustration, $c = 20, \ell = 5$ is covered whenever even $n \ge 22$.

This counterexample uses a prescribed nonzero-age initialization. It does not apply to mandatory partner removal or duplicate-processing rules that refresh stored ages. The next proposition addresses reachability from zero ages for one parameter choice, while carefully distinguishing pathwise and distributional statements.

Proposition 3 (A frozen state reachable from zero ages)**.** *Under retention, $n = 4, c = 2, \ell = 1$, independent uniform initiators, uniform age-tie resolution, and uniform responder subsets, there is a strongly connected full-cache, zero-age initialization from which a frozen identity state is reached with probability at least 2^{-24}. Consequently almost-sure time-averaged uniform sampling fails for that initialization.*

Proof. Start from

$$C_1 = \{2^0, 4^0\}, \quad C_2 = \{3^0, 4^0\}, \quad C_3 = \{1^0, 2^0\}, \quad C_4 = \{1^0, 2^0\}.$$

Superscripts denote ages. This graph is strongly connected. Execute the following triples of initiator, selected partner, and returned identity:

$$(2,4,1),\ (1,4,2),\ (3,1,2),\ (3,1,4),\ (4,2,1),\ (4,2,3).$$

For $\ell = 1$, an accepted incoming descriptor replaces the single sent entry; a filtered descriptor makes no replacement. Direct substitution in these rules gives

$$C_1 = \{4^1, 3^0\}, \quad C_2 = \{3^1, 4^0\},$$
$$C_3 = \{2^2, 4^1\}, \quad C_4 = \{1^2, 3^1\}.$$

The pairs $(1,4)$ and $(2,3)$ now satisfy the frozen-state condition, with common sets $\{3\}$ and $\{4\}$ respectively. Each of the six specified initiators has probability $1/4$, each specified oldest-partner choice is between two tied entries, and each specified response has probability $1/2$. The exact prefix probability is therefore $(1/16)^6 = 2^{-24}$. On this event, peer 1 never again samples peer 2, so its empirical frequency for that identity tends to zero rather than $1/3$. □

Positive-probability absorption by itself does *not* prove that the unconditional local marginal law fails to approach uniform from the zero-age initialization: different frozen outcomes could balance in their mixture. Theorem 2 directly disproves that marginal-law guarantee when the specified nonzero initial ages are allowed.

4 An availability obstruction under mandatory removal

Now choose initiators independently and uniformly, remove the partner unconditionally, and let an activation of an empty cache do nothing.

Proposition 4 (Persistent failure of full-capacity requests)**.** *Let $c = n - 1$ and $1 \le \ell \le c$. For every peer i and $t \ge 1$, even starting from complete caches,*

$$\Pr\{S_i^{(c)}(t) = \perp\} \ge \frac{1}{n}, \qquad \left\| \mathcal{L}(S_i^{(c)}(t)) - \mathrm{Unif}((V \setminus \{i\})_c) \right\|_{\mathrm{TV}} \ge \frac{1}{n}.$$

Proof. If activation t selects i, either its cache is already empty or it removes some partner q. Since q's cache contains no self entry q, its response cannot restore q to i. Thus i has at most $n-2 < c$ entries at the end of that activation. The event that i was selected has probability $1/n$. The distance bound follows by testing the event $\{\perp\}$, which has target probability zero. □

This concerns unconditional $b = c$ requests. It does not establish nonuniformity for smaller samples conditional on availability. Retrying or conditioning changes the sampling model.

5 A positive capacity-one theorem

5.1 Precisely specified variant

Set $c = \ell = 1$, retain the current neighbor when a reply is unusable, and choose the initiator independently and uniformly at each step. With $q = f(p)$ and $r = f(q)$, the complete identity update is

$$T_p f = \begin{cases} f, & r = p, \\ f \text{ with } f(p) \leftarrow r,\ f(q) \leftarrow p, & r \neq p. \end{cases} \tag{3}$$

Initial ages may be arbitrary: the single cached entry is necessarily oldest, so ages have no effect on this identity projection. Let Ω_n be the nonself functions $f : V \to V$ whose underlying undirected graph is connected, and let

$$P(f,g) = \frac{1}{n} \#\{p : T_p f = g\}.$$

Theorem 5 (Convergence of the capacity-one retention variant)**.** *For every $n \geq 2$, $f \in \Omega_n$, and observer i,*

$$\mathcal{L}_f(f_t(i)) \longrightarrow \mathrm{Unif}(V \setminus \{i\}).$$

For $n \geq 4$, with $D = 2n(n-2)$ and $\delta = n^{-D}$,

$$\|\mathcal{L}_f(f_t(i)) - \mathrm{Unif}(V \setminus \{i\})\|_{\mathrm{TV}} \leq (1-\delta)^{\lfloor t/D \rfloor}. \tag{4}$$

Consequently $D\lceil n^D \log(1/\varepsilon)\rceil$ activations suffice for distance at most ε. For $n = 3$, $(8/9)^{\lfloor t/2 \rfloor}$ is sufficient. For $n = 2$, the sample is already uniform.

This is an extremely weak quantitative bound and gives no efficient mixing claim. The theorem concerns identity marginals, not stationarity of the complete age process.

5.2 Proof for $n \geq 4$

Lemma 6 (Inverse moves and connectivity)**.** *Ω_n is closed under T_p. Every nontrivial move initiated by p with $q = f(p)$ is inverted by initiating q in the resulting state. In particular P is symmetric.*

Proof. A nontrivial move replaces a path on the three distinct vertices p, q, r by another path on the same vertices, preserving undirected connectivity and avoiding self entries. In the resulting state $g(q) = p, g(p) = r$; initiating q restores $f(q) = r, f(p) = q$. This pairs each marked transition (f, p, g) with its unique reverse marked transition (g, q, f), so the transition counts agree. □

Lemma 7 (Insertion into a cycle)**.** *Every state in Ω_n reaches a directed Hamiltonian cycle in at most $n-2$ moves.*

Proof. A connected functional digraph has exactly one directed cycle: outgoing paths eventually reach a cycle, and an edge cannot join basins of distinct cycles. The cycle length is at least two. While an off-cycle vertex remains, choose p whose parent $q = f(p)$ lies on the cycle. Initiating p replaces its cycle edge $q \to r$ by $q \to p \to r$, adding one vertex to the cycle. □

Lemma 8 (Adjacent swap)**.** *If a Hamiltonian cycle has consecutive distinct vertices x, a, b, y, the activation word x, x, a, b swaps a, b, leaving every other outgoing edge unchanged.*

Proof. The only changing outgoing edges are

	$f(x)$	$f(a)$	$f(b)$
initial	a	b	y
x	b	x	y
x, x	y	x	x
x, x, a	a	y	x
x, x, a, b	b	y	a

All moves are valid for $n \geq 4$. At $n = 4$ the temporary residual cycle $x \leftrightarrow y$ is allowed. □

Lemma 9 (Diameter)**.** *For $n \geq 4$, the transition graph on Ω_n is connected with diameter at most $D = 2n(n-2)$.*

Proof. Reach a Hamiltonian cycle from each of two states in at most $n-2$ steps. Anchor the cyclic notation at one fixed vertex. The other $n-1$ vertices form a linear ordering; any two orders are related by at most $(n-1)(n-2)/2$ adjacent swaps. Each costs four moves by Lemma 8. Reverse the second insertion path using Lemma 6. The total is at most $2(n-2)+2(n-1)(n-2) = 2n(n-2)$. □

Proof of Theorem 5 for $n \geq 4$. The uniform law π_n on Ω_n is stationary because P is symmetric. Choose the state $z(1) = 2, z(2) = 1, z(j) = 1$ for $j \geq 3$. Initiating 1 at z is a self-loop. Every state reaches z along a path of at most D moves. Pad shorter paths with that self-loop to exactly D activations. The prescribed sequence has probability $n^{-D} = \delta$, so

$$P^D(f, \cdot) \geq \delta\, \delta_z(\cdot) \quad (f \in \Omega_n).$$

Writing $P^D(f, \cdot) = \delta\, \delta_z + (1 - \delta)R(f, \cdot)$ yields

$$\|\mu P^D - \nu P^D\|_{\mathrm{TV}} \leq (1 - \delta)\|\mu - \nu\|_{\mathrm{TV}}$$

for any probability laws μ, ν. Iterate with $\nu = \pi_n$, then use total-variation contraction for the remaining steps:

$$\|P^t(f, \cdot) - \pi_n\|_{\mathrm{TV}} \leq (1 - \delta)^{\lfloor t/D \rfloor}.$$

Relabelling bijects Ω_n onto itself and preserves π_n. Permutations fixing i act transitively on the other identities, so $\pi_n\{f : f(i) = j\} = 1/(n - 1)$ for each $j \neq i$. Projection gives (4). The time bound follows from $(1 - \delta)^k \leq e^{-\delta k}$. □

5.3 Small network sizes

For $n = 2$ the sole state is the reciprocal pair and its local target distribution has one element. For $n = 3$ the identity chain has two four-state classes. One consists of

$$H = (2, 3, 1), \quad L_1 = (3, 1, 1), \quad L_2 = (2, 1, 2), \quad L_3 = (3, 3, 2).$$

The other is obtained by relabelling a reversed triangle. The center H moves to each leaf with probability $1/3$. Each leaf returns to H with probability $1/3$ and stays with probability $2/3$. These classes contain all eight nonself maps. Uniform law on each class is stationary, and every observer points to either other identity in exactly two of its four states.

Choose a leaf $L_\star$ in the appropriate class. From every state, $P^2(f, L_\star) \geq 1/9$: from another leaf go through H; from H use the target leaf's self-loop; from the target leaf use two self-loops. The same minorization argument proves the stated bound. This also shows why irreducibility of the full identity space is stronger than necessary for local uniformity.

6 A general symmetry reduction, not a mixing proof

Lemma 10 (Symmetrized reference process)**.** *Let a Markov transition kernel commute with peer relabelling, and suppose the sample extraction rule is equivariant and always defined. Fix initial state x, observer i, and the group H_i of permutations fixing i. Start a reference process Y_t from*

$$\bar{\mu}_x = \frac{1}{|H_i|} \sum_{\sigma \in H_i} \delta_{\sigma x}.$$

Then $S_i^{(b)}(Y_t)$ is exactly uniform on the ordered distinct b-tuples of other peers at every time. For any coupling,

$$\|\mathcal{L}_x(S_i^{(b)}(X_t)) - \mathrm{Unif}((V \setminus \{i\})_b)\|_{\mathrm{TV}} \leq \Pr\{S_i^{(b)}(X_t) \neq S_i^{(b)}(Y_t)\}.$$

Proof. The initial law of Y is invariant under H_i. Equivariance preserves this invariance at every time and transfers it to the sample law. Since H_i acts transitively on the output tuples, that law must be uniform. The remaining claim is the coupling inequality. □

This lemma requires no finite state space, stationary age law, or full-state irreducibility. If samples can be unavailable, the uniformity conclusion holds only conditional on availability. The lemma provides no forgetting bound by itself. In Theorem 2, its reference law is uniform while the original local law remains permanently nonuniform, so the required disagreement probability cannot tend to zero.

7 Exact model for the mandatory-removal results

The positive results that follow use the enhanced exchange in the published CYCLON paper [2, Section 2.2], with the following explicit completion of its choices. Membership is fixed, exchanges are reliable and atomic, and $1 \le \ell \le c \le n-1$. Each cache C_p contains at most c distinct nonself identities, each with a finite nonnegative integer age.

At each global event, choose an initiator independently and uniformly from V. An empty initiator does nothing. Otherwise, the initiator p increments all its cached ages and selects a maximum-age partner q, breaking ties uniformly. It selects $\min(\ell-1, |C_p|-1)$ other entries uniformly. Write S for these other entries with their incremented ages. Its outgoing buffer is $\{p^0\} \cup S$, and it unconditionally removes q from its cache. The responder chooses a uniform subset of size $\min(\ell, |C_q|)$ of its *pre-exchange* cache and returns it. Both sides discard incoming self entries and incoming identities already present locally, without refreshing duplicate ages. They insert every remaining incoming entry, using vacancies first and then replacing sent entries still resident locally. Every eligible choice of victims has positive probability, independent of identities. For uniform quantitative bounds one may choose uniformly among the eligible sets. Entries not selected as victims remain in place.

The two buffers are formed before either cache processes incoming entries. Removing q creates a vacancy at p; it does not remove the other sent entries. At q, all sent entries are initially resident. These rules always provide enough eligible slots. If the pre-exchange cache sizes are a, b, the initiator has $c - a + 1 + \min(\ell-1, a-1) \ge \ell$ vacancies plus eligible residents, and the responder has $c - b + \min(\ell, b) \ge \ell$. The incoming buffer size at either side is at most ℓ.

This defines a countable-state Markov chain including ages. Independent equal-rate Poisson activation clocks give this same embedded chain. Fixed periodic phases, which the original paper uses, give a different scheduler. Thus none of the positive results below silently asserts convergence for every periodic schedule. The underfull-buffer convention and uniform tie-breaking above are also mathematical specifications, rather than claims that the paper uniquely prescribes them.

8 From full-state convergence to local uniformity

The following facts isolate what a general proof must establish. A Markov kernel is *equivariant* if relabelling peers in a state and then taking a step has the same law as taking a step and relabelling its outcome.

Lemma 11 (A common atom gives convergence, including ages)**.** *Let P be a Markov kernel on a countable state space. Suppose there are $L \ge 1$, $\delta > 0$, and a state z such that*

$$P^L(x, \{z\}) \ge \delta \qquad \text{for every state } x.$$

Then P has a unique stationary probability law π, and

$$\sup_x \|P^t(x,\cdot) - \pi\|_{\mathrm{TV}} \le (1-\delta)^{\lfloor t/L \rfloor}.$$

Irreducibility of the whole state space is not required.

Proof. The case $\delta = 1$ is immediate. Otherwise split $P^L(x,\cdot) = \delta\delta_z + (1-\delta)R(x,\cdot)$, where R is a Markov kernel. Hence $\mu \mapsto \mu P^L$ is a contraction by $1-\delta$ in total variation. Probability measures on a countable space are complete in this metric, so the contraction has a unique fixed point π. Since $(\pi P)P^L = (\pi P^L)P = \pi P$, uniqueness gives $\pi P = \pi$. Every P-stationary law is also P^L-stationary, proving uniqueness for P. Iterating the contraction and applying the remaining $t \bmod L$ transitions proves the bound. □

Theorem 12 (Stationary symmetry and availability)**.** *Let P be equivariant on a relabelling-invariant state space. Suppose it has a unique stationary law π and $\eta_t(x) = \|P^t(x,\cdot) - \pi\|_{\mathrm{TV}} \to 0$. Fix i and $1 \le b \le c$, and suppose $a = \pi\{|C_i| \ge b\} > 0$. For uniform sampling without replacement from C_i, followed by uniform ordering,*

$$\mathcal{L}_\pi(S_i^{(b)} \mid |C_i| \ge b) = \mathrm{Unif}((V \setminus \{i\})_b).$$

The same conditional sample law converges to this uniform law from every initial state x. If $\eta_t(x) \le a/2$, its distance is at most $4\eta_t(x)/a$.

Proof. For every permutation σ of V, equivariance makes $\sigma_*\pi$ stationary. Uniqueness gives $\sigma_*\pi = \pi$. The event $A = \{|C_i| \ge b\}$ is invariant under permutations fixing i, and the output rule is equivariant under this group. Its action on distinct ordered b-tuples of other identities is transitive. An invariant probability law on that finite set is uniform.

For the quantitative claim include the extra randomness of sample extraction in the transition from a state to $(A, S_i^{(b)})$. Total variation cannot increase under this transition. Put $a_t = \Pr_x(A \text{ at } t)$. Then $|a_t - a| \le \eta_t(x)$. For any set B of available tuples,

$$\left| \frac{\Pr_x(A, S_i^{(b)} \in B)}{a_t} - \frac{\Pr_\pi(A, S_i^{(b)} \in B)}{a} \right| \le \frac{2\eta_t(x)}{a_t} \le \frac{4\eta_t(x)}{a}.$$

Taking the supremum over B proves the result. □

At stationarity the *unconditional* output law, including the unavailable symbol, is $(1-a)\delta_\perp + a\,\mathrm{Unif}((V \setminus \{i\})_b)$. Its distance from the always-available uniform target is exactly $1-a$. Conditioning at a fixed observation time is the sampling rule in Theorem 12. Waiting or retrying until a cache can answer introduces a stopping-time rule and requires a separate argument. Nor does this theorem assert independent samples across peers or time, or a uniform distribution on global overlay graphs.

9 A general age-reset result for shuffle length one

Theorem 13 (Finite-set regeneration for $\ell = 1$)**.** *Use the mandatory-removal model of Section 7 with arbitrary n, c and $\ell = 1$. Label the peers $1, \dots, n$ and set*

$$K = n + c - 1, \qquad L = nK, \qquad \delta = (nc)^{-L}.$$

Let $\mathcal{F}$ be the finite set of states with all ages zero and $C_i \subseteq \{i+1, \dots, n\}$ for every i. Then, from every state with arbitrary finite ages,

$$P^L(x, \mathcal{F}) \ge \delta.$$

If $\tau_{\mathcal{F}}$ is its first hitting time, then

$$\Pr_x\{\tau_{\mathcal{F}} > mL\} \le (1-\delta)^m, \qquad \mathbb{E}_x \tau_{\mathcal{F}} \le L/\delta.$$

Proof. First activate one fixed peer p for K consecutive events. Only its cache ages are incremented during this block. When its chosen partner q already knows p, prescribe that q return the descriptor for p. This occurs with probability at least $1/c$. The reply is a self entry at p, so mandatory removal reduces $|C_p|$ by one. The request contains only p^0, a duplicate at q; hence q retains its descriptor for p.

If q does not know p, its receipt of p^0 makes it learn p. It either uses a vacancy or replaces its single returned entry. Subsequent exchanges in this block cannot make it forget p: it never

initiates, and every request from p contains only the already-known identity p. There are at most $n-1$ such first-learning events. The size of C_p never increases, because one descriptor is removed and at most one is received. Every other nonempty activation in the prescribed block shrinks it. At most c shrinkings empty it. Consequently $n-1+c=K$ activations suffice, independently of all ages. Remaining activations of an empty p do nothing.

Now execute this block for $p=1,2,\dots,n$ in order. Immediately after its block, p is empty. During later blocks it never initiates, and every incoming request contains only a fresh self descriptor j^0 from a later peer $j>p$. Thus its final cache consists solely of age-zero descriptors of later peers. The final state belongs to $\mathcal{F}$.

Prescribing the L initiators has probability n^{-L}. Each reply that must be prescribed has conditional probability at least $1/c$; there are at most L of them. No particular oldest-partner or victim choice was required. The resulting lower bound is $(nc)^{-L}$. Applying it to successive L-event blocks, using the Markov property, gives the tail bound, and summing that bound gives the expectation. □

This proves that unbounded initial ages can be forgotten often enough to hit a finite family. It does not prove that the different members of $\mathcal{F}$ communicate or have identical limiting local marginals. In particular, $\mathcal{F}$ need not have a unique empty sink. Weak connectivity is preserved for $\ell=1$: the contacted edge is retained in reverse, and if the responder deletes its sole returned edge, that edge is transferred to the initiator or is already present there. Hence the same hitting statement holds for $\mathcal{F}$ intersected with the weakly connected states when that is the initial state space. The common-atom construction in Section 11 will connect this family for arbitrary n, c. First we record a general criterion showing what the age reset alone implies.

9.1 Convergence without assuming a unique recurrent class

Lemma 14 (A finite reset family with self-loops)**.** *Let P be a countable-state Markov kernel. Suppose a finite nonempty family F, an integer $L\ge 1$, and $\delta>0$ satisfy*

$$P^L(x,F)\ge\delta \quad \text{for every } x, \qquad P(f,\{f\})>0 \quad \text{for every } f\in F.$$

Then there are finitely many closed communicating classes $C_1,\dots,C_m$, with $m\le|F|$, each having a unique stationary probability law π_k. Every initial state enters their union almost surely. There exist $B\ge 1$ and $0<\alpha<1$ such that, writing $h_k(x)$ for the probability of eventually entering C_k,

$$\sup_x \left\| P^t(x,\cdot)-\sum_{k=1}^{m} h_k(x)\pi_k \right\|_{\mathrm{TV}} \le 2(1-\alpha)^{\lfloor t/(2B)\rfloor}. \tag{5}$$

The weights sum to one. This conclusion permits the limiting stationary law to depend on the initial state.

Proof. On the finite vertex set F, draw $f\rightsquigarrow g$ when some finite positive-probability path of the full chain goes from f to g. Consider the terminal strongly connected components of this reachability graph and choose an atom z_k in each. Every $f\in F$ reaches at least one such atom.

Let C_k be the set of all full-chain states reachable from z_k. If $y\in C_k$, the reset bound gives a path from y to some $g\in F$. Terminalness implies that g and z_k lie in the same strongly connected component of the finite graph. Hence y can return to z_k. Thus C_k is a communicating class, and it is closed because every state reachable from one of its states is reachable from z_k. Different terminal components give disjoint classes.

For each $f\in F$, prescribe a finite positive-probability path to one selected $z_{k(f)}$. Since F is finite, the maximum path length M is finite. Pad all paths to length M using the self-loop

at their terminal atoms. Let $\varepsilon > 0$ be the minimum probability of these finitely many padded paths. Set $B = L + M$ and choose $0 < \alpha \le \min(\delta\varepsilon, 1/2)$. Then, with $C = \bigcup_k C_k$,

$$P^B(x, C) \ge \alpha \quad \text{for every } x.$$

Moreover, if $x \in C_k$, its reset states lie in $F \cap C_k$, and their prescribed paths necessarily lead to z_k. Therefore

$$P^B(x, \{z_k\}) \ge \alpha \quad (x \in C_k).$$

Lemma 11, applied to the closed space C_k, gives its unique stationary law and convergence bound $(1-\alpha)^{\lfloor t/B \rfloor}$. The first inequality and closedness give $\Pr_x(\tau_C > s) \le (1-\alpha)^{\lfloor s/B \rfloor}$. In particular $\sum_k h_k(x) = 1$.

Split $t = s + r$, where $s = \lfloor t/2 \rfloor$. Conditional on $X_s \in C_k$, evolve for r more steps and compare with π_k. On $X_s \notin C$, compare the remaining evolution with the mixture of its eventual class laws; this contribution is bounded by the probability of that event. The Markov property gives $\mathbb{E}_x h_k(X_s) = h_k(x)$, so the comparison law is exactly $\sum_k h_k(x)\pi_k$. Hence the total distance is at most

$$(1-\alpha)^{\lfloor s/B \rfloor} + (1-\alpha)^{\lfloor r/B \rfloor} \le 2(1-\alpha)^{\lfloor t/(2B) \rfloor}.$$

All closed communicating classes must intersect F by the reset bound, and so are among the classes just constructed. □

Theorem 15 (Arbitrary-capacity convergence for singleton shuffles)**.** *For every $n \ge 2$ and $1 \le c \le n-1$, mandatory-removal CYCLON with $\ell = 1$ and independent uniform activations converges in full-state total variation from every weakly connected initial state with arbitrary finite integer ages. Its limit has the form $\sum_k h_k(x)\pi_k$ in Lemma 14, and the uniform geometric bound (5) holds for some constants depending on n, c and the specified transition rule.*

Proof. Restrict to the closed weakly connected state space. Theorem 13 supplies its finite reset family $F = \mathcal{F} \cap \{\text{weakly connected states}\}$ and a uniform reset probability. Every $f \in F$ has $C_n = \varnothing$, so activating n gives a self-loop of probability $1/n$. Apply Lemma 14. □

This is a general convergence statement for the age-and-cache process. By itself it does not prove forgetting of the initial configuration or uniform local samples. The constants B, α above are existential: their construction uses finite paths between the reset states, without an explicit bound on those path lengths. To invoke Theorem 12 one must still establish a unique stationary regime, or prove that all possible limiting regimes and their initial-state-dependent weights give the desired uniform local law. The next two sections resolve this issue for every capacity and provide explicit convergence bounds.

10 A complete mandatory-removal theorem at capacity one

Theorem 16 (Every network size, capacity one)**.** *Let $n \ge 2$ and $c = \ell = 1$ in Section 7. On the weakly connected state space, the full cache-and-age chain has a unique stationary law π: the uniform law on all labelled rooted spanning trees, with all descriptor ages zero and all arcs directed toward the empty root. Set $D = 2n(n-1)$ and $\beta = n^{-D}$. Then*

$$\sup_x \|P^t(x, \cdot) - \pi\|_{\mathrm{TV}} \le (1-\beta)^{\lfloor t/D \rfloor}.$$

At stationarity, every observer i has

$$\pi(C_i = \varnothing) = \frac{1}{n}, \qquad \pi(C_i = \{j^0\}) = \frac{1}{n} \quad (j \ne i).$$

Thus the sample converges to uniform on the other peers conditional on a nonempty cache, with limiting availability $1 - 1/n$.

Proof. Represent a singleton cache by its identity $f(p)$ and an empty cache by $f(p) = \perp$. If p is nonempty, put $q = f(p)$ and $r = f(q)$. The identity update is

$$f'(q) = p, \qquad f'(p) = \begin{cases} \perp, & r \in \{p, \perp\}, \\ r, & r \notin \{p, \perp\}. \end{cases}$$

Other pointers do not change. A path on three distinct vertices $p \to q \to r$ becomes $q \to p \to r$. Contact with an empty responder reverses the edge. A reciprocal exchange removes only one of its two opposite arcs. Weak connectivity is therefore preserved.

A weakly connected digraph of outdegree at most one consists either of a directed cycle with trees feeding into it, or of a rooted spanning tree with one empty root. Indeed, following outgoing pointers reaches a cycle or an empty vertex, and different terminal cycles or empty vertices cannot have intersecting weak basins. On a directed cycle of length $k \geq 3$, activating a cycle vertex shortens that cycle by one. On a two-cycle, activating either endpoint produces a rooted tree. At most $n - 1$ prescribed activations thus reach a rooted tree, which remains a rooted tree.

In a rooted tree the initiator's descriptor is deleted, the responder's descriptor transfers to the initiator if present, and the responder receives a fresh age-zero descriptor. There is no copying, and the only incremented descriptor is deleted. Consequently the number of positive-age descriptors decreases by one whenever a positive-age carrier initiates, and never increases. At most $n - 1$ further prescribed activations clear every positive age. The zero-age rooted-tree family is closed.

We next prove that its identity transition graph is connected. Fix an anchor a and follow its directed path to the root. While an off-path vertex remains, choose one, p, whose parent q is on the path. Activating p inserts p immediately after q on the path; if q is the root, p becomes the new root. Such a vertex exists by connectivity of the tree. At most $n - 1$ moves give a Hamiltonian path beginning at a.

On a Hamiltonian path with consecutive vertices x, u, v, write $y = f(v)$, allowing $y = \perp$ when v is the root. The word x, x, u, v swaps u, v and changes no other outgoing edge:

	$f(x)$	$f(u)$	$f(v)$
initial	u	v	y
x	v	x	y
x, x	y	x	x
x, x, u	u	y	x
x, x, u, v	v	y	u

All four moves are valid, including when $y = \perp$. Adjacent swaps sort the $n-1$ vertices following a, using at most $(n-1)(n-2)/2$ swaps. Every nontrivial tree move initiated by p is inverted by initiating its old parent q in the resulting tree, including when that parent was the empty root. Therefore we may reverse the insertion path of a target tree. The tree transition graph has diameter at most

$$2(n-1) + 4\,\frac{(n-1)(n-2)}{2} = 2(n-1)^2.$$

Choose one fixed zero-age rooted star z. The cycle reduction, age clearing, and tree diameter bound give a path to z of length at most $2(n-1)+2(n-1)^2 = D$ from every initial aged state. Pad to exactly D events using the empty root's no-op. Every prescribed initiator has probability $1/n$; there are no nontrivial buffer or tie choices at capacity one. Hence $P^D(x, \{z\}) \geq n^{-D}$. Lemma 11 gives uniqueness and the displayed bound.

Pairing each marked tree transition (f, p, g) with its inverse (g, q, f) proves symmetry of the identity kernel on the finite tree family. Its uniform law, with all ages zero, is therefore stationary and must equal π. Relabelling makes the root uniform on V. Permutations fixing i make the remaining $n-1$ possible cached identities equiprobable, each with mass $(1-1/n)/(n-1) = 1/n$. Conditional convergence follows from Theorem 12. □

11 A complete arbitrary-capacity theorem for singleton shuffles

We now close the remaining common-regime gap for $\ell = 1$. For an ordered set $V = \{1, \dots, n\}$, define $H(V, c)$ by giving peer i the $\min(c, n-i)$ greatest labels larger than i, all at age zero. Its largest vertex is empty, and every other vertex points to that largest vertex.

Theorem 17 (Arbitrary population and cache capacity, $\ell = 1$)**.** *Let $n \geq 2$ and $2 \leq c \leq n-1$ in Section 7, with $\ell = 1$. Define, using this fixed capacity c,*

$$B_1 = 0, \qquad B_m = 2m(m + c - 1) + 2B_{m-1} \quad (2 \leq m \leq n),$$

$$T = n(n + c - 1) + B_n, \qquad \gamma = (nc^2)^{-T}.$$

From every weakly connected state with arbitrary finite integer ages,

$$P^T(x, \{H(V, c)\}) \geq \gamma.$$

The full cache-and-age chain therefore has a unique stationary law π and

$$\sup_x \|P^t(x, \cdot) - \pi\|_{\mathrm{TV}} \leq (1 - \gamma)^{\lfloor t/T \rfloor}.$$

For every observer and every $1 \leq b \leq c$, availability has a positive stationary probability, and the local ordered b-sample converges to uniform conditional on availability. Together with Theorem 16, this covers every allowed capacity for shuffle length one.

11.1 Clearing blocks and separating one source

Write $\mathcal{B}(p)$ for the clearing block in Theorem 13: activate only p; whenever its partner already knows p, prescribe reply p, and otherwise choose any reply. With m active vertices it empties p within $m + c - 1$ events. No unselected resident of the initiator can be replaced when $\ell = 1$, since the sole usable reply fits the vacancy made by partner removal. If all ages are zero at the block's start, they are zero at its end: only p increments ages, and its cache is empty at the end; passive peers receive only fresh p^0.

Lemma 18 (An unknown source with a connected remainder)**.** *From any zero-age weakly connected state on V, clear its smallest vertex a, then clear each vertex of $W = V \setminus \{a\}$, keeping a passive. At the end all ages are zero, no cache in W contains a, C_a is nonempty, and the graph induced by W is weakly connected.*

Proof. Immediately after clearing a, every component of the induced graph on W has a descriptor for a: C_a is empty while the whole graph is connected. During the remaining blocks the component partition on W can only coarsen. A W-internal exchange preserves its internal connectivity by edge reversal and transfer; if its reply is a, it deletes no W edge. An exchange with passive a likewise deletes no W edge and may add one through the reply.

The first contact with a inserts its first cached identity and defines an anchor component in W. Thereafter every cached identity at a belongs to that anchor, which may grow. Indeed a new initiator p contacting a receives an anchor identity, joining the anchor, or receives itself and was already in it. A nonempty cache at passive a cannot become empty: inserting a new initiator either fills a vacancy or replaces one entry, while a duplicate changes nothing.

Every initial component joins the anchor. A descriptor for a can disappear either when its holder contacts a, joining the anchor directly, or when a full responder returns it and replaces it by the active peer's fresh identity. In the latter case it transfers to, or is already present at, that active peer. The reversed edge links the two within W, and its clearing block must eventually contact a. Thus eliminating a component's last such descriptor forces that component into the anchor.

At the end no W cache contains a: each vertex was emptied and later passive requests contain only other initiators' identities. Consequently all components have joined, and at least one contact has made C_a nonempty. Since a is now globally unknown, it cannot reappear while a stays inactive. All block boundaries have zero ages. □

11.2 Extending the canonical remainder

Lemma 19 (Setting the cache of an unknown source)**.** *Let $c \geq 2$. Suppose all ages are zero, the smallest vertex a is unknown in $W = V \setminus \{a\}$, the induced state on W is $H(W, c)$, and C_a is any nonempty cache in W. There is a sequence of $|V|$ clearing blocks ending with all ages zero, a still unknown in W, W weakly connected, and C_a equal to the $\min(c, |W|)$ greatest members of W.*

Proof. Let $m = \max W$, $d = \min(c, |W|)$, let J be the d greatest members of W, and put $S = J \setminus \{m\}$.

First clear a, using reply a from a responder that knows a, and otherwise its smallest cached identity. Every W vertex other than m retains its edge to m. A full responder has at least two entries, so its smallest entry is not m; an underfull responder evicts nothing. The block must visit m. Otherwise let q be the greatest visited vertex. At its first visit its untouched canonical cache returns a label greater than q. That label either is already pending at a or is inserted, and cannot disappear without being visited, contradicting maximality of q. Since m initially is empty and remains passive, this phase ends with $C_m = \{a^0\}$ and $C_a = \varnothing$. The remainder stays connected through its edges to m.

Next, for each $j \in S$, start its clearing block with $j \to m$ and prescribe reply a. Preserve the descriptor for m at every unprocessed vertex: at a full responder not knowing j, choose any reply other than m; at one knowing j, reply j. Underfull responders need not evict anything. Thus each unprocessed j still knows m when its block begins. All ages are zero, so its first choice of m is a legal tie. Before that contact C_m consists of a and the previously processed members of S, with size at most $d - 1 < c$. Fresh j therefore fills a vacancy and preserves a there. On later visits, m already knows j and replies j. The active j obtains or retains a descriptor for a, so must contact a before its block ends. Passive a has fewer than c entries during this phase. After these blocks,

$$C_a = S, \qquad C_m = \{a\} \cup S,$$

all at age zero, and every processed j is free of a.

Now clear all $h \in W \setminus J$. Keep a at passive m: if m is full and does not know h, choose a reply other than a, possible because $c \geq 2$; otherwise use the ordinary clearing policy. At passive a, protect every member of S. If this phase is nonempty, $|S| = c - 1$, so there is at most one unprotected slot. Fill a vacancy for a new h, or return and replace the unique unprotected entry. For a duplicate h, reply h. These choices respect the clearing-block rule. At the end the only remaining descriptor for a is at m: every other vertex was cleared after its last possible acquisition of a, and later fresh requests cannot reintroduce that identity. Cache C_a consists of S and at most one unprotected entry.

Finally start m's clearing block with $m \to a$, a zero-age tie. At a, reply with and replace its unprotected entry, if present; otherwise fill its vacancy. Fresh m is new there, since m has never initiated in these phases. Hence C_a becomes exactly J. This removes the globally last descriptor for a. Finish clearing m; it cannot contact a again, so C_a stays fixed. All ages are zero at the end.

After the first phase W was connected. Thereafter a is passive; exchanges with it only add W edges, while internal exchanges preserve W connectivity. Thus W remains connected. The number of clearing blocks is $1 + |S| + |W \setminus J| + 1 = |V|$. □

11.3 Induction and the convergence bound

Proof of Theorem 17. For the induction allow a fixed configured capacity $c \geq 2$ even on a smaller active population. Excess cache slots cause no difficulty; the cache is still a set of nonself identities from that population. The one-vertex case is empty.

On a zero-age connected m-vertex state, apply Lemma 18, then freeze its unknown smallest vertex a. Use induction to make W canonical. Nothing can contact a, so its cache stays fixed. Apply Lemma 19 to set C_a to the greatest $\min(c, |W|)$ labels, leaving a unknown, W connected, and all ages zero. Freeze a again and make W canonical a second time. This reaches exactly $H(V, c)$. The two lemmas use m clearing blocks each, of at most $m + c - 1$ events, and the two recursive calls cost at most $2B_{m-1}$. Thus the stated recurrence bounds the entire path.

From arbitrary ages first apply the clearing sweep of Theorem 13, costing $n(n+c-1)$ events and leaving a connected zero-age state. The total path length is at most T. Pad shorter paths at the empty largest vertex of $H(V, c)$. At each prescribed event the initiator has probability $1/n$, the chosen oldest tie has probability at least $1/c$, and the prescribed nonempty reply has probability at least $1/c$. Empty replies and no-ops require fewer choices. Therefore the path probability is at least $(nc^2)^{-T}$. Lemma 11 proves the full-state conclusion.

For any observer, relabel the same construction so that it is the smallest vertex of the canonical state and therefore has c entries. Stationarity and the same minorization imply that its full-cache probability is at least γ. The transition rule and connected state space are equivariant, so Theorem 12 proves conditional uniformity for every $1 \leq b \leq c$. □

Proposition 20 (A robustness consequence)**.** *Let P_1 be the singleton-shuffle kernel in Theorem 17. Suppose another equivariant kernel Q preserves the same weakly connected aged state space and satisfies $Q(x, A) \geq \varepsilon P_1(x, A)$ for every state x and event A, for a fixed $\varepsilon > 0$. Then Q also converges to a unique stationary law, with conditional uniform local samples, and has the bound*

$$\sup_x \|Q^t(x, \cdot) - \pi_Q\|_{\mathrm{TV}} \leq (1 - \varepsilon^T \gamma)^{\lfloor t/T \rfloor}.$$

Proof. Kernel domination gives $Q^T \geq \varepsilon^T P_1^T$. Use the same common atom and then Lemmas 11 and Theorem 12. Relabelled canonical atoms again give positive availability. □

For example, this covers a protocol that independently chooses a singleton shuffle with probability ε and otherwise takes any equivariant step preserving the connected state space. It is a corollary for that explicitly randomized-length protocol; it does not assert domination for fixed $\ell > 1$ CYCLON.

12 A complete multi-entry mandatory-removal theorem

Theorem 21 (Three peers with two cache entries)**.** *In the mandatory-removal model of Section 7, let $n = 3$ and $c = \ell = 2$. Start from any weakly connected identity overlay, with arbitrary finite nonnegative integer ages. The complete cache-and-age chain has a unique stationary law π on this state space, and*

$$\sup_x \|P^t(x, \cdot) - \pi\|_{\mathrm{TV}} \leq (1 - 6^{-20})^{\lfloor t/20 \rfloor}. \tag{6}$$

For every observer and $b = 1, 2$, the stationary probability of availability is positive and the conditional local sample converges to $\mathrm{Unif}((V \setminus \{i\})_b)$.

The bound is deliberately crude. This is a proof for actual oldest-first, age-carrying exchanges with multi-entry caches, but it does not establish a theorem for arbitrary n, c, ℓ. At $n = 3$ the conditional $b = 2$ output is automatically uniform when both possible neighbors are present; convergence and positive availability still require proof.

12.1 Three-peer transition rules

Write q^a for a descriptor of identity q and age a. A full cache contains both possible other identities. The responder never needs to delete an entry: it simply adds any new nonself identity, since only two such identities exist. A full initiator ends with exactly its other, uncontacted entry, whose age was incremented. It cannot receive the contacted identity from that identity's own cache. A singleton initiator ends with the responder's third identity, if present, and is otherwise empty.

The weakly connected state space is closed. If a full cache is present after an exchange, all three vertices are connected. Otherwise an exchange with an empty responder reverses an edge, and a reciprocal singleton exchange deletes only one of the two reciprocal arcs. A path through distinct singleton initiator and responder creates a full responder. These cases cover every exchange and preserve weak connectivity.

12.2 Resetting arbitrary ages on a chain

For three distinct role names F, S, T, call the state a *chain* if

$$C_F = \{S^a, T^b\}, \qquad C_S = \{F^d\}, \qquad C_T = \{S^e\}. \tag{7}$$

Define the zero-age state $Z(F, S, E)$ by

$$C_F = \{S^0, E^0\}, \qquad C_S = \{E^0\}, \qquad C_E = \varnothing.$$

Lemma 22 (Chain age reset)**.** *Every chain reaches a state of form Z in at most eight exchanges, regardless of its finite initial ages.*

Proof. Suppose first $b \geq a$. Prescribe the following five exchanges:

$$F \to T, \quad S \to F, \quad F \to S, \quad T \to S, \quad T \to F.$$

The complete trace is

after	C_F	C_S	C_T
initial	$\{S^a, T^b\}$	$\{F^d\}$	$\{S^e\}$
$F \to T$	$\{S^{a+1}\}$	$\{F^d\}$	$\{S^e, F^0\}$
$S \to F$	$\{S^{a+1}\}$	$\varnothing$	$\{S^e, F^0\}$
$F \to S$	$\varnothing$	$\{F^0\}$	$\{S^e, F^0\}$
$T \to S$	$\varnothing$	$\{F^0, T^0\}$	$\{F^1\}$
$T \to F$	$\{T^0\}$	$\{F^0, T^0\}$	$\varnothing$

The fourth choice is legal because the two ages at T after its increment are $e + 1$ and 1; S is oldest, possibly tied when $e = 0$. All other partner choices are forced by singleton caches, apart from the first, which is allowed by $b \geq a$. The final state is $Z(S, F, T)$.

If instead $a > b$, activate F toward S. This produces a chain with roles (S, T, F) and full-cache ages $(b + 1, d)$ for its near and far vertices, respectively. If $d \geq b + 1$, the preceding reset applies. Otherwise integrality gives $d \leq b$; rotate again by activating the new full vertex toward its near vertex. The roles are now (T, F, S) and the full ages are $(d + 1, e)$. If $e \geq d + 1$, reset. Otherwise $e \leq d \leq b$; a third rotation gives roles (F, S, T) with full ages $(e + 1, b + 1)$. Now the far vertex is oldest, and the five-exchange reset applies. At most three rotations and five further exchanges are needed. □

12.3 Reaching the chain topology

Lemma 23 (Identity reduction compatible with oldest-first choices)**.** *Every weakly connected three-peer state reaches a chain in at most seven exchanges.*

Proof. Classify states by the numbers of full, singleton, and empty caches. All named activations below use an oldest entry; where it matters, the age inequalities are given explicitly.

Two full caches and one singleton. Let A, B be full and C singleton. Activate A. If it contacts B, there is now one full cache and two singletons. Otherwise it contacts C, which becomes full, and A becomes singleton. If C previously pointed to B, its full cache has ages B^u, A^0 for some $u \geq 0$. Activate C toward the full vertex B; this is oldest or tied and reduces to one full cache. If C previously pointed to A, its cache is $\{A^u, B^{v+1}\}$, where v was the age of B at A. If B is oldest, again activate C toward B. Otherwise activate C toward the singleton A. Then A is full with cache $\{B^{v+1}, C^0\}$. It can contact B, its strictly oldest entry. Thus at most three exchanges reach one full cache and two singletons.

Other states with at least two full caches. Three full caches become two full and one singleton after one activation. With two full and one empty, activate a full cache. It either contacts the other full cache, yielding one full, one singleton, and one empty, or contacts the empty vertex, yielding two full caches and a singleton. Use the preceding case thereafter.

One full cache and two singletons. If the singleton arrows form a chain ending at the full vertex, the required form already holds. If both point to the full vertex, activate either singleton; it now points to the other singleton, giving a chain. The remaining case has reciprocal singleton arrows. Let Q be an oldest neighbor of the full vertex F and R the other vertex. Activate $Q \to R$, emptying Q, then $F \to Q$. The resulting state is a chain with full vertex Q. At most two exchanges are needed.

One full F, one singleton S, and one empty E. If $S \to F$, first activate it, giving $S \to E$. If the oldest neighbor of F is S, activate $S \to E$, emptying S; this does not change the ages at F. Now activate F toward an empty oldest neighbor. This gives one full cache and two singletons, handled above. There are at most three preliminary exchanges.

One full and two empty. One full activation gives the preceding pattern.

No full caches. With three singletons, weak connectivity implies a directed path $P \to Q \to R$ on distinct vertices. Activate $P \to Q$; Q becomes full and the other caches are singleton. With two singletons and one empty, there is either a path toward the empty root or two arrows into that root. In the path case, activate the leaf toward its singleton parent, producing one full, one singleton, and one empty. In the two-arrow case, reverse one arrow by activating it toward the empty root, then use the path case. This takes at most two preliminary exchanges. There are no other weakly connected size patterns.

The longest route just described is the last pattern: at most two exchanges to one full, one singleton, and one empty; at most three to one full and two singletons; and at most two to a chain. It costs seven. The two-full/empty route costs at most $1+3+2=6$, the three-full route at most $1+3+2=6$, and the other routes are no longer. □

12.4 A common zero-age state

Lemma 24 (Relabelling the zero-age roles by legal exchanges)**.** *Every $Z(F,S,E)$ reaches every prescribed labelled state of form Z in at most four exchanges.*

Proof. The exchange $S \to E$ maps $Z(F,S,E)$ to $Z(F,E,S)$ in one step. The two exchanges $F \to S$ followed by $F \to E$ map $Z(F,S,E)$ to $Z(S,E,F)$. The first choice is an age-zero tie; before the second exchange the remaining entry of F has age one. Direct substitution shows that both final states have all ages zero.

If the desired full vertex is currently S, use the two-step rotation. If it is currently E, first swap S, E and then rotate, costing three steps. If it is already F, do nothing. One final swap, if needed, sets the two remaining roles. □

Proof of Theorem 21. Lemmas 23 and 22 reach some zero-age Z in at most $7+8=15$ exchanges. Lemma 24 reaches the fixed state $z = Z(1,2,3)$ within four more. Pad the resulting path to

exactly twenty activations by repeatedly activating the empty vertex at z. Every prescribed initiation has probability $1/3$ and every prescribed oldest choice has probability at least $1/2$. The buffers are deterministic in this parameter setting, and the responder never chooses victims. Therefore

$$P^{20}(x, \{z\}) \geq 6^{-20} \quad \text{for every weakly connected } x.$$

Lemma 11 proves existence, uniqueness, and (6), without truncating ages or assuming a finite state space.

The same construction works for any relabelled z. Integrating its minorization against π shows $\pi(\{z'\}) \geq 6^{-20}$ for each labelled Z state z'. For each observer choose one with that observer full. Thus $\pi\{|C_i| \geq b\} \geq 6^{-20} > 0$ for $b = 1, 2$. Theorem 12 gives the conditional local conclusion. □

13 Controlled progress for general shuffle lengths

The next lemma addresses one difficulty caused by larger buffers: copying a newly created descriptor can evict another such descriptor. It concerns a prescribed legal exchange, rather than the average behavior of the random process.

Attach an auxiliary color, old or fresh, to each descriptor. Copies inherit their source color, and every new age-zero self descriptor is fresh. Discarding a duplicate does not change the stored descriptor's color. Colors are proof bookkeeping and do not change the protocol or its oldest-first selection.

Lemma 25 (A controlled fresh-count inequality)**.** *Use Section 7 with arbitrary n, c, ℓ. Suppose a nonempty initiator p selects an old partner descriptor q. For every prescribed extra initiator buffer S, there exist a responder buffer and eligible victim choices for which the global number N_f of stored fresh descriptors does not decrease. It increases by at least one if the responder accepts any fresh incoming descriptor. All these choices have positive probability under uniform buffer and eligible-victim selection.*

Proof. Let $P = C_p$ and $Q = C_q$ denote the pre-exchange identity sets, $a = |P|$, $b = |Q|$, $k = |S| = \min(\ell - 1, a - 1)$, and $m = \min(\ell, b)$. Let S_f be the fresh descriptors in S, using their identities when taking intersections. The responder must insert

$$J = (\{p\} \cup S) \setminus Q, \qquad j = |J|.$$

Of these, $f_J = 1_{\{p \notin Q\}} + |S_f \setminus Q|$ are fresh. There are $v_Q = c - b$ vacancies, so $r_Q = (j - v_Q)_+$ entries must be replaced.

Call a descriptor in Q protected if it is fresh and its identity lies in $P \cup \{p\}$. Put $d_0 = |Q \cap (\{p\} \cup S)|$. Since $q \notin Q$, at most $d_0 + (a - 1 - k)$ descriptors are protected. Thus the number of nonprotected entries is at least

$$b - d_0 - a + 1 + k \geq k + 1 - d_0 - (c - b) = j - v_Q.$$

Choose r_Q nonprotected victims D. Choose the responder buffer T to include these victims and the entries with identities

$$B = (S_f \cap Q) \cup (\{p\} \cap Q).$$

This is possible: the union lies in Q, and if $r_Q > 0$ its size is at most $d_0 + r_Q = k + 1 - v_Q \leq \ell$. If $r_Q = 0$, its size is at most $k + 1 \leq \ell$. Complete T arbitrarily to size m. Fill vacancies and then replace exactly D. Every fresh victim at the responder has identity absent from $P \cup \{p\}$; its fresh copy in T is therefore accepted by the initiator.

The initiator accepts $I = T \setminus (P \cup \{p\})$. Write $i = |I|$, f_I for its fresh count, $f_S = |S_f|$, $d = |S_f \cap Q|$, and $h = 1_{\{p \in Q\}}$. Because T includes B, $i \le m - d - h$. Deleting the old partner leaves $v_P = c - a + 1$ vacancies. Choose old extra-buffer victims before fresh ones. The number of fresh descriptors the initiator must remove is at most

$$\big(i - v_P - (k - f_S)\big)_+ \le (f_S - d - h)_+ = (f_J - 1)_+.$$

Here $m - v_P \le k$ follows from both $m - v_P \le \ell - 1$ and $m - v_P \le a - 1$. The responder loses at most f_I fresh descriptors, all of which have fresh copies accepted at the initiator. Combining the two endpoints gives

$$\Delta N_f \ge f_J - (f_J - 1)_+ = 1_{\{f_J > 0\}}.$$

This proves the controlled inequality and its strict version. □

This is neither a pathwise monotonicity theorem for every random exchange nor an expected-drift inequality. It does not handle an activation that removes a fresh partner. It also does not show that a strict increase can always be arranged within a bounded number of activations. That strict-progress problem remains a gap.

There is a limited bridge to numerical ages: over H global events, any descriptor lineage gains at most H age units, and copying preserves its age. Let A be the maximum of zero and all initial fresh-descriptor ages. If every initially old descriptor has age greater than $A + H$, an initiator containing an old descriptor selects an old partner throughout this horizon; new fresh descriptors start at age zero. A bounded colored clearing strategy compatible with actual oldest choices would therefore help construct an age reset. Such a strategy has not been proved here for arbitrary parameters.

14 Numerical age control and initial-lineage erasure

Lemma 26 (Eviction under consecutive activations)**.** *Use Section 7 with arbitrary n, c, ℓ. Suppose a descriptor r of age a is present at p. Let M_s be the maximum initial age of identity s anywhere in the network, using zero if it is absent, and let B be the initial maximum age. If only p is activated, the resident descriptor r is removed, either as partner or as a replacement victim, within*

$$U = 1 + \sum_{s \notin \{p,r\}} (M_s - a + 1)_+ \le 1 + (n-2)(B+1)$$

activations. If the replacement choices retain r until it is selected, it is selected within this bound. In particular, an initially globally maximum-age descriptor cannot remain resident through $n-1$ consecutive activations of its holder.

Proof. Let t count completed activations of p, and put $K_s(t) = M_s(t) - t$, with $M_s(t)$ the current global maximum age of identity s, again zero if absent. For $s \ne p$, K_s never increases: only p increments ages, copies retain their ages, and no new self descriptor of identity s is born. When s is selected as partner, its sole descriptor at p is deleted and is absent from the outgoing buffer. Responder s cannot return its own identity. Every other descriptor of identity s retains its age, so K_s decreases by at least one.

While the original r remains resident, its age after t activations is $a + t$. A different partner s selected next must therefore satisfy $K_s(t) \ge a$. Each s can be selected at most $(M_s - a + 1)_+$ times before r is removed. Summing these counts proves the first bound. If $a = B$, every summand is at most one, giving $U \le n - 1$. □

Lemma 27 (Protecting one resident descriptor)**.** *If $\ell < c$, suitable positive-probability buffer and victim choices can retain any one designated nonpartner descriptor at the initiator and any one designated descriptor at the responder. If $c = n - 1$, no nonpartner descriptor is ever replaced, for any $\ell \le c$.*

Proof. For an initiating cache of size $a > \ell$, choose the extra buffer to exclude its protected entry. If $a \le \ell$, all $a-1$ nonpartner entries are sent, but mandatory removal creates $c-a+1 \ge \ell-a+2$ vacancies. At most ℓ usable replies require at most $a-2$ replacements, leaving room to preserve one of those $a-1$ entries.

For a responder of size $b > \ell$, exclude its protected entry from its reply buffer. If $b \le \ell$, it sends all entries, but $c-b \ge \ell+1-b$ vacancies permit at most $b-1$ replacements. These buffer and victim choices can be made simultaneously at the two endpoints.

For $c = n-1$, any cache union with usable nonself incoming identities has size at most c. At the initiator the removed partner cannot be returned by that partner. Thus all usable incoming identities fit vacancies, without replacing a resident. □

Lemma 28 (Following a target through forced replacements)**.** *For every $1 \le \ell \le c$, a descriptor of identity r and age a can be followed through a positive-probability sequence until its current holder selects r as partner. If the initial global maximum age is B, this takes at most*

$$1 + \sum_{s \ne r} (M_s - a + 1)_+ \le n(B+1)$$

events, where M_s is the initial global maximum age of identity s, with value zero if absent. The followed copy gains exactly one age unit at every event preceding its deletion.

Proof. Always activate the current holder p. If $\ell < c$, retain the tracked nonpartner by Lemma 27. Consider the remaining case $\ell = c$, when the outgoing buffer contains every nonpartner resident. Retain the target whenever possible. If this is impossible, every one of the $a_p - 1$ nonpartner residents must be replaced, where a_p is the current cache size. Since mandatory deletion creates $c - a_p + 1$ vacancies, the number i of usable replies must satisfy $i-(c-a_p+1) = a_p-1$, hence $i = c$. The responder's full cache is consequently disjoint from the initiator's pre-exchange cache and self identity. In particular it does not know r. The outgoing tracked copy of r is accepted there, with its incremented age. Follow it.

For arbitrary changing holders, put $K_s(t) = M_s(t) - t$ for every identity, taking its maximum age as zero if absent. This quantity never increases: ages gain at most one per event, copies preserve ages, and fresh self descriptors start at zero. When s is selected as partner, all other descriptors of identity s keep their ages, while the incremented one is removed and cannot be copied or returned by responder s. Thus K_s decreases by at least one. Until the tracked r is selected, its age is $a + t$. Each different selected identity s must have $K_s(t) \ge a$, and can be selected at most $(M_s - a + 1)_+$ times. Sum over all $s \ne r$; unlike Lemma 26, the holder changes, so no fixed holder identity can be excluded. □

Theorem 29 (Accessibility of initial-lineage erasure, all parameters)**.** *For every $1 \le \ell \le c \le n-1$, from every state of Section 7 with finite maximum age M_0, there is a finite positive-probability execution that erases every initially present descriptor and all descendants obtained by copying it. With $D_0 = 2n(n-1)$, the execution can be chosen to have length at most*

$$T_0 = ((n+1)^{D_0} - 1)(M_0 + 1).$$

Every descriptor remaining at its end was generated during this execution and has age at most its length. Connectivity is not needed for this accessibility statement.

Proof. Mark all initial descriptors old, let their copies inherit the mark, and mark every newly generated self descriptor fresh. We eliminate the old descriptors one identity at a time. Fix an identity r. Within its stage, keep a growing set of immunized peers, each storing a fresh descriptor for r. Keep immunized peers passive and preserve their fresh r. For $\ell < c$ this is possible by Lemma 27. For $\ell = c$, whenever a received buffer contains an identity already

known at its responder, there are at most $c-1$ usable incoming entries. A responder of size b then replaces at most $(c-1-(c-b))_+ \le b-1$ residents, permitting protection of one.

Start with any old copy of r and apply the moving-target construction of Lemma 28. No immunized peer is chosen as a holder. In the full-shuffle case, every nonterminal outgoing packet contains the tracked old r. If its responder is immunized, that entry is a duplicate, allowing preservation of the fresh resident r. Moreover such a responder replies with its own fresh r, which is a duplicate at the old holder. It gives at most $c-1$ usable replies, so cannot force migration of the tracked copy into that immunized responder.

When r is finally selected by holder q, that holder loses its descriptor for r, and responder r cannot return it. The outgoing fresh self descriptor ensures that r stores q afterward. If q was already known, its duplicate leaves at most $c-1$ other usable incoming entries in the full-shuffle case, so its existing descriptor can be protected. If new, the inserted descriptor is not an eligible victim in that exchange.

Now activate only the source peer r, keeping its cached q until it is selected or until retention is impossible. The eviction bound of Lemma 26 makes this block finite. If q is selected, it has remained passive and free of r, so accepts fresh r and becomes immunized. If q is instead unavoidably evicted, the proof of Lemma 28 shows that the responder has c usable replies and is missing identity r. It accepts the outgoing fresh self descriptor of r and becomes immunized. Stop the block at this event. Either outcome adds a new immunized peer. Every previously immunized responder sees the duplicate fresh self descriptor r, so can preserve its resident fresh r even under full shuffling.

Each iteration adds a distinct immunized peer, and none reacquires an old r. After at most $n-1$ iterations all old descriptors for r are gone. Extinction is irreversible: copying preserves identity and new self descriptors are fresh. Repeat for all identities, using at most D_0 routing or single-source forcing blocks.

For the length bound let S be elapsed execution length and put $R = M_0+S+1$. Current ages are at most M_0+S. Each block has length at most nR by the two numerical-age lemmas, so it multiplies R by at most $n+1$. After at most D_0 blocks, $M_0+S+1 \le (n+1)^{D_0}(M_0+1)$, proving the bound. Every remaining descriptor starts at age zero during this execution and gains at most one per event. All prescribed choices have positive probability, proving accessibility. □

The bound depends on M_0; this is not a uniform finite-set regeneration or a recurrence theorem. Erasing initial descriptor lineages alone does not erase the influence of the initial topology on the locations of the fresh descriptors.

14.1 Maximal capacity and connectivity

Proposition 30 (Monotone connectivity at maximal capacity)**.** *For $c = n-1$ and every $1 \le \ell \le c$, the underlying undirected graph never loses an edge. From every weakly connected state with finite ages, a complete underlying graph is accessible by a finite positive-probability path. Any stationary probability law on this state space, if one exists, is supported on complete underlying graphs.*

Proof. By Lemma 27, the only removed resident descriptor is the mandatory partner at the initiator. Its undirected edge is preserved by the reverse descriptor at the responder. All other existing edges persist.

If the connected underlying graph is not complete, it has an induced two-edge path $u-v-w$ with u, w nonadjacent. Lemma 26 permits forcing contact with any currently cached target by activating only its holder; no nonpartner can be replaced in this regime. If the orientation contains a directed path $u \to v \to w$, force u to contact v and include w in the reply. This adds the missing edge, unless it was already added along the way. If $u \to v \leftarrow w$, first force $u \to v$, creating $v \to u$, then force $w \to v$ and reply with u. If $u \leftarrow v \to w$, activate only

v until its first contact with either endpoint. The other descriptor remains resident, and the contacted endpoint learns v. Freeze v and use the resulting directed path as in the first case. Selecting the desired reply entry has positive probability for every $\ell \geq 1$. These cases cover the possible orientations, including bidirectional edges. At most two forcing blocks add an edge. Monotonicity allows repetition until the graph is complete.

Finally, edge count is bounded and nondecreasing. Under a stationary law it cannot increase with positive probability over any finite horizon. Every noncomplete state has a finite positive-probability path increasing it, so cannot have stationary mass. The state space is countable, which proves the support claim. □

This proposition proves accessibility, rather than almost-sure completion or existence of a stationary law. The forcing path lengths may depend on ages. For full shuffling $\ell = c = n - 1$, there is also the exact knowledge-set relation

$$K'_q = K_p \cup K_q, \qquad K'_p = (K_p \cup K_q) \setminus \{q\}, \qquad K_i = C_i \cup \{i\}.$$

A bookkeeping token moved to each responder accumulates knowledge monotonically. Turning this into a uniform age-reset construction remains an open step here.

15 Established results and remaining problem

The shuffle-length-one case is now proved for every population size and cache capacity: Theorems 16 and 17 establish full-state convergence from arbitrary finite ages, unique stationarity, positive availability, and conditional local uniformity. The latter proof supplies the missing common-atom construction: source separation, a reserved callback, and induction connect every zero-age state to the same labelled canonical overlay. At capacity one the stationary law is explicit; at higher capacities symmetry identifies the local conditional law without requiring an explicit formula for the global stationary law. Theorem 21 additionally handles $n = 3, c = \ell = 2$. The earlier retention theorem concerns a different convention.

For every shuffle length, Theorem 29 now proves accessibility of erasing all initial descriptor lineages, including the diagonal regime $\ell = c < n - 1$. Numerical age routing and either a selected callback or a forced replacement produce a new protected fresh copy at each iteration. The construction still has an age-dependent length, and fresh descriptor locations can retain initial-topology information. Thus recurrence and a common limiting regime for fixed $\ell \geq 2$ remain unproved in general. The precise replacement rule must also preserve the proposed state space before a stationary symmetry argument is applied to it.

A sufficient next step is an age-independent reset bound for fixed $\ell \geq 2$, or a suitable recurrence argument followed by a common-regime proof. A labelled common atom would give convergence by Lemma 11 and conditional uniformity by Theorem 12, just as in the singleton case. Alternatively, Proposition 20 already proves convergence for connected-state-preserving protocols that occasionally perform singleton shuffles with uniformly positive probability. This is an explicit protocol extension rather than a proof for every fixed shuffle length.

The scheduling and sampling target cannot be omitted. Theorem 1 rules out a theorem covering every fixed periodic phase schedule at $c = \ell = 1$. The retention-specific frozen states in Theorem 2 do not refute the mandatory-removal target. Proposition 4 concerns unconditional full-capacity requests and does not refute conditional uniformity. The present note does not prove the arbitrary-parameter CYCLON convergence claim, nor any efficient or polylogarithmic mixing bound.

16 Finite verification

The accompanying `verify.py` checks the periodic trace including ages, every possible initiating/subset choice in a small frozen-twin example, the zero-age absorption certificate, the frozen construction's connectivity, the availability argument on small cache states, and the capacity-one identity chain exhaustively for $2 \le n \le 6$. It checks inverse moves, communicating classes, stationary local counts, and Hamiltonian adjacent swaps. It covers 14,831 connected identity states, 87,858 marked transitions, and 564 Hamiltonian swap words.

The additional checker `verify_reset.py`, using `cyclon3.py`, both in `mandatory-removal/`, follows the constructive three-peer reset without truncating the ages reached along its paths. It exhaustively checks all 46,550 weakly connected initial states whose stored ages belong to $\{0, 1, 2, 3, 4\}$. Every constructed transition is checked against the exact oldest-first kernel; every path reaches $Z(1, 2, 3)$ within the proved nineteen-exchange limit. The longest observed path in this finite check has fourteen exchanges. The symbolic proof above, rather than this finite enumeration, establishes the theorem for arbitrarily large finite integer ages. All these checks corroborate the proofs; none replaces a missing general convergence argument.

The checker `verify_capacity_one.py` in the same directory checks all weakly connected capacity-one identity states through $n = 6$, including empty caches, cycle reduction, tree closure, inverse moves, tree irreducibility, uniform local counts, and the Hamiltonian-path swap gadget, including swaps next to the empty root.

The checker `verify_singleton_general.py` checks the entire arbitrary-capacity induction with exact ages, including every oldest choice, clearing bound, protected cache entry, unknown-source invariant, induced connectivity requirement, and zero-age block boundary. It passes 1,102 exhaustive small forward states and 264 random aged initializations through $n = 10$. The largest constructed trajectory in those checks has 8,472 events. An independent implementation additionally passed 6,030 exhaustive connected zero-age states through $n = 4$ and 1,000 random arbitrary-age initializations through $n = 9$, with ages up to 9,999.

The checker `verify_lineage_erasure.py` checks 3,300 initial states covering every parameter combination through $n = 10$, including empty and underfull caches and disconnected states. It checks legal buffers, replacement victims, numerical routing bounds, immune-peer preservation, and complete initial lineage extinction. Its runs exercise 307 forced migrations and have maximum observed length 2,237 events. These finite and random checks support the symbolic proofs; the all-parameter lineage result remains an accessibility theorem, not a general mixing theorem.

The complete mathematical text is contained in this single `main.tex`; it has no external TeX inputs or bibliography file.

References

[1] R. Guerraoui, A.-M. Kermarrec, A. Kucherenko, R. Pinot, and M. de Vos. *PEERSWAP: A Peer-Sampler with Randomness Guarantees*. arXiv:2408.03829, 2024. https://arxiv.org/pdf/2408.03829.

[2] S. Voulgaris, D. Gavidia, and M. van Steen. *CYCLON: Inexpensive Membership Management for Unstructured P2P Overlays*. Journal of Network and Systems Management, 13:197–217, 2005. Published article, author-hosted copy: https://acropolis.aueb.gr/~spyros/www/papers/Cyclon.pdf.